

Enhanced Two-Photon Excited Emission of Quantum Emitters by Colloidal Plasmonic Metasurfaces

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Colloidal plasmonic metasurfaces are emerging as versatile platforms for advancing nanophotonic applications, whose rational design of increasing complexity allows for tailoring and integrating advanced plasmonic and photonic optical properties. Traditional lithographic techniques limit crystallographic and chemical adaptability, hindering the dynamic tuning of optical properties. In this work, a template-assisted self-assembly approach for achieving plasmonic metasurfaces using colloidal silver nanoparticle arrays and a thin film of core-shell CdSe/ZnS quantum dots are demonstrated. This system leverages plasmonic surface lattice resonances to create open nanophotonic cavities that enhance the two-photon excited emission process. The near-field interactions within these nanophotonic structures offer spectral tunability and efficient light-matter coupling. Moreover, using synthetic nanoparticles enables the integration of a rich library in nanoparticles' size, shape, and composition, positioning them as promising candidates for compact and scalable photonic devices targeting nonlinear optical processes, advanced sensing, and dynamic light manipulation.

to the Rayleigh–Wood (R–W) anomalies, where the diffracted modes will transition from evanescent to propagating.^[1] The hybridization of the R–W's anomalies (characterized by a sharp spectral feature) and the plasmonic response of each unit composing the array leads to the formation of collective optical responses, known as surface lattice resonances (SLRs).^[2–4] The resulting SLRs exhibit strong EM confinements, intense extinction cross sections, high spatial delocalization, longer lifetimes, and narrow bandwidths that can be exploited to promote various nonlinear optical phenomena.^[5–7] Moreover, the hybridization will also have important consequences on the optical tunability of the obtained collective resonances, showing a strong dependency on the illumination angle, as well as on the associated electric field distribution that is not confined to

1. Introduction

When arranging plasmonic nanoparticles (NP) into ordered arrays with a period (Λ) comparable to the wavelength of the incident electromagnetic (EM) radiation, unique collective optical responses arise. The localized surface plasmon resonances (LSPR) of each NP in the array can couple with the diffractive orders generated by the incident light impinging on a patterned surface, which acts as a plasmonic grating at wavelengths close

to the interparticle region, resulting in a reduction of the optical losses.^[8]

Traditionally, such photonic materials are fabricated by electron beam lithographic templates, followed by a metal deposition step under high vacuum. For example, recent studies have shown that using a layer with a high refractive index to match the array can induce the formation of waveguiding modes that hybridize SLRs. By varying the thickness of the high-refractive-index layer, a significant effect on the optical response is observed, which can be used to enhance the emission properties of gain media.^[9] Indeed, employing index matching layers offers the opportunity to compensate for the refractive index mismatch between superstrate and substrate and observe the far-field optical response even in lower quality structures. However, this process inevitably seals the metasurface and makes it inaccessible. As a result, these conventional fabrication approaches inherently offer limited control over both the crystallography and the surface chemistry of the prepared metasurfaces. These two aspects are fundamental for the development of dynamic metamaterials capable of interacting with the chemical surroundings, driving new applications in biosensing,^[10,11] catalysis,^[12,13] and wearable devices.^[14,15] In this scenario, fabrication using colloidal building blocks effectively combines the undisputed control over crystal structure and surface chemistry achieved by synthetic nanochemistry with the scalability and versatility of colloidal self-assembly strategies.^[16–25]

Exploiting the wide library of building blocks, the resulting plasmonic metasurfaces are characterized by a higher flexibility in design, enabling multiple LSPRs and SLRs to be placed at

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specific wavelengths across the visible–infrared spectra.^[26] These advantages thrust a significant effort toward the development of colloidal self-assembly as a high-throughput approach for implementing colloidal plasmonic arrays on a variety of different materials.^[27,28] As such, colloidal metasurfaces emerged as a valid alternative for the development of chemically active plasmonic nanophotonic cavities and metamaterials, encouraging more multidisciplinary exploration and opening new research pathways.^[26,29] For example, templated self-assembly of silver NP have been rationally designed to promote both normal and off-normal lasing emission using a completely open and accessible optical cavity.^[30]

In the context of nanoplasmonics, the last decade has witnessed a significant effort towards the improvement of two-photon (2P) excited fluorescence efficiency in plasmon–emitter-coupled systems,^[31] exploiting fluorescent dye probes with larger 2P absorption cross sections, and the rational design of metal NPs capable of confining the exciting EM field in the proximity of the gain media, ultimately enhancing fluorescence intensity, shortening fluorescence lifetime, and improving photostability.^[32–34] Although colloidal plasmonic NPs have proven to be particularly advantageous for various applications in near-field enhanced spectroscopy, there are only very few examples in the literature describing the use of colloidal-based plasmonic metasurfaces for the study of high-order optical processes.^[34–36] However, several challenges stemming from both practical and theoretical points of view are still preventing significant progress for the application of colloidal plasmonic building blocks for these types of systems.^[36–39]

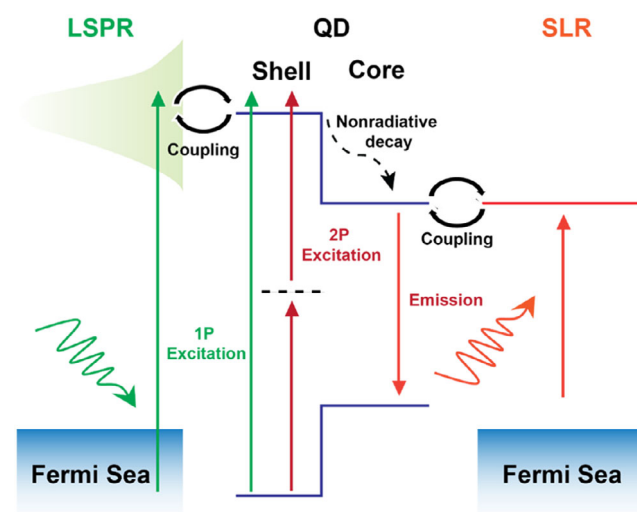
In this work, we report on the enhancement and spectral tunability of 2P-excited emission from core–shell CdSe/ZnS quantum dots (QDs) using colloidal-based plasmonic metasurfaces fabricated via templated self-assembly as an open optical cavity. The first-order SLR was tuned to match the emission wavelength for the QDs and promote an enhanced 2P-excited emission with a near-infrared (NIR) excitation source. The near-field enhanced 2P photoluminescence also showed the signature spectral tunability on the excitation angle typically associated with SLR. Moreover, the unique open-cavity structure of the waveguide metamaterial enables the interrogation of the fundamental insights of the near-field enhancement mechanism. Our findings underscore the potential of these types of metamaterials for the future development of more compact and efficient photonic devices, crucial for the next generation of high-speed, high-bandwidth optical networks.

2. Results

2P excited emission, one of the most commonly studied third-order nonlinear optical phenomena, consists of a quantum emitter simultaneously absorbing two photons with identical energy to an electronic state and consequently emitting a single higher-energy photon.^[40–42] Unlike one-photon (1P) excitation, where fluorescence can be excited along the entire beam path, 2P excitation leaves fluorophores outside the focal plane unexcited, significantly improving spatial selectivity and reducing background noise. The sharp spatial confinement of the excitation region minimizes fluorescence from surrounding areas, improving

contrast and signal detection sensitivity. Indeed, under 2P excitation, photobleaching is minimized outside the focal plane, preserving sample integrity. In nonlinear optics, this technique enables the study of multiphoton interactions, second-harmonic generation, and quantum effects in photonic materials.^[43] Moreover, the characteristic 2P excitation small excitation volumes make it an ideal technique for high-resolution microscopy, allowing for deep-tissue imaging with minimal photobleaching and background fluorescence, improving contrast and resolution.^[44] In this work, we set to study the 2P excited emission properties resulting from a plasmonic square array consisting of colloidal silver NPs coupled with a thin layer of close-packed CdSe/ZnS core–shell QDs. The choice of an inorganic nanocrystal emitter ensures emission spectra tunability, high quantum efficiency, and exceptional photostability, preventing undesirable photobleaching issues often generated by the high pump power required to promote nonlinear optical phenomena. The transitions and possible couplings under 1P and 2P excitation are summarized in **Scheme 1**. The 1P excitation (515 nm) excites both the LSPR and the QDs, and the emission is enhanced by the SLR, while under 2P excitation (1030 nm), only the SLR is coupled to the QDs.

The QDs used in this work have an average diameter of 9 ± 1 nm, and once dispersed in a toluene solution, they present a broad absorption peak that extends over 550 nm, an emission maximum at 622 nm, and a quantum efficiency of 50% (Figure S1, Supporting Information). Before integrating the QDs, a thin film of high refractive index photoresist (SU8) with a thickness of ≈ 250 nm was spin coated over a glass substrate (**Figure 1a** and Table S1, Supporting Information). Recent studies have shown that the use of a high-refractive-index layer on the plasmonic array can induce the formation of waveguiding modes that hybridize the SLRs.^[9,30] In the proposed architecture, the presence of an SU8 layer induces a refractive index mismatch between substrate and superstrate (air), inducing the formation of such a hybrid waveguided-lattice resonance, ultimately enabling narrow-bandwidth optical response arising from



Scheme 1. The energy diagram shows transitions and couplings of the LSPR, SLR, and quantum-dot excitons under 1P and 2P excitation.

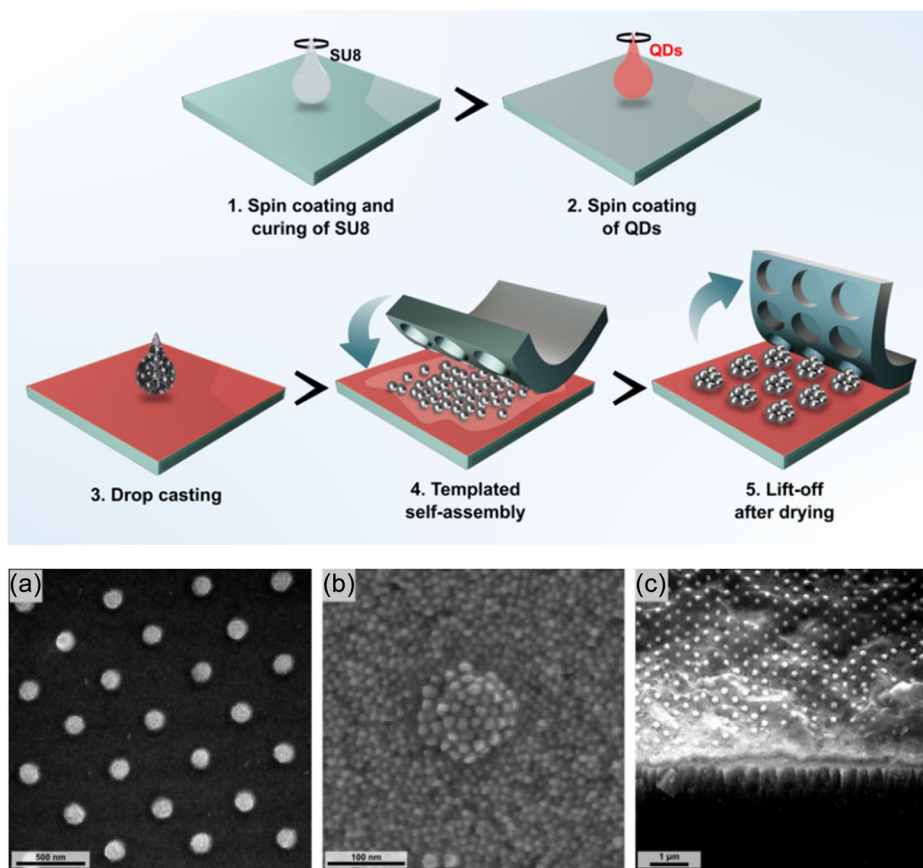


Figure 1. Top: Fabrication scheme for the 2D squared array of spherical AgNPs with core-shell QDs (CdSe/ZnS) used for 2P excited emissions. **Bottom:** SEM images of: a) array ($\Lambda = 400$ nm) of AgNPs on top of a layer of QDs. b) High-magnification SEM image of the AgNPs cluster on top of a layer of core-shell type of QDs. c) SEM cross section (45°) of the array assembled on top of a layer of QDs deposited onto the SU8. Additional SEM images can be found in the Supporting Information (Figure S4).

completely exposed optical cavities.^[30] Next, the dispersed QDs were spin coated on top of the cured SU8 to form a thin layer used as the substrate for the self-assembly of the plasmonic metasurface. Since the metallic nanostructures can alter the total quantum efficiency of the system by altering the radiative and nonradiative decay rates,^[45] the spectral correlation between lattice modes of plasmonic nanostructures and the absorption/emission spectra of the gain media is crucial, due to the strongly enhanced local excitation field. In this work, the metasurfaces were prepared starting from silver NP (AgNPs) with an average diameter of 18 ± 2 nm (Figure S2, Supporting Information), arranged in arrays of clusters with lattice parameter (Λ) of 400 nm (Figure 1, details on AgNPs synthesis, concentration, absorption, and ligand exchange can be found in the **Experimental Section**). The choice of Ag is motivated by its low optical losses in the visible portion of the EM spectrum compared to other noble metals such as gold. Nonetheless, plasmonic resonances remain affected by intrinsic damping mechanisms mainly associated with the conversion of plasmonic energy into heat rather than radiative emission, reducing the overall efficiency of SLR-mediated processes, especially for nonlinear optical applications. For this reason, the dimension of the silver colloids was specifically selected in order to minimize the

near-field coupling within the repeating cluster unit that builds up the arrays, ultimately reducing the optical losses of the system to a minimum and maximizing the quality factors of the fabricated optical cavity. This effect has been reported in previous studies, demonstrating how the use of small nanoparticle building blocks (< 20 nm) is a convenient strategy to reduce the optical losses associated with near-field coupling within the system and to maximize the large-scale uniformity of the sample.^[17,46,47] Although sharp resonances can be obtained by single-particle plasmonic arrays,^[48] from a templated colloidal self-assembly standpoint, the use of small AgNPs is a more straightforward method for creating resonances with high quality factors at the expense of possible structural defects, which can impact the lattice response of single NP plasmonic arrays.^[49] Overall, the final architecture has been rationally designed to keep the photonic cavity completely exposed, and the lattice period of 400 nm was chosen to exploit the high-refractive-index layer (QDs/SU8) and generate a lattice resonance that can overlap with the emission of the QDs. Before the assembly, the silver colloids were coated with poly(ethylene glycol) methyl ether thiol (PEG-SH) with an average molecular weight of 2000 g mol^{-1} ; this ligand shell allows the steric stabilization of the AgNPs throughout the assembly process, enabling to form densely close-packed

assemblies,^[17,50] and confer long-term colloidal stability and resistance toward oxidation of the environment exposed cavities, with no decrease in performance that could be registered over a period of 12 months (Figure S3, Supporting Information). Overall, substrate preparation could be finalized in less than 1 h, and the metasurfaces are ready for characterization as soon as the self-assembly is complete (typically 4–8 h). Scanning electron microscopy (SEM) analysis shows excellent large-scale assembly, and high-magnification imaging clearly reveals the presence of the QDs underneath the AgNPs (Figure 1b,c). The thickness of the QDs layer was evaluated by cross-section SEM (Figure 1c) as well as atomic force microscopy (AFM) (Figure S5, Supporting Information), confirming an average thickness of ≈ 10 nm, which corresponds to a single layer of QDs.

First, the transmission profile of the resulting metasurface was characterized by illuminating the sample along the normal direction (Figure 2). The broad transmission band between 500 and 600 nm can be attributed to the clusters' LSPR modes, a direct result of near-field coupling between the close-packed plasmonic colloids.^[17] In addition to the localized plasmonic response, the 2D arrangement of the AgNPs strongly modulates the optical properties of the system, as can be seen by comparing the normal incidence transmission measured for the sample without array (red curve in Figure 2a), and with array (blue and black curves): the sharp angle-dependent transmission band at 611 nm is indicative of the presence of an SLR mode. This narrow bandwidth resonance ($Q_F = 34$)^[51] is a direct consequence of the formation of the waveguided lattice modes within the SU8 layer, avoiding the need for a superstrate index-matching layer, leaving an open-cavity system. Another important conclusion to be drawn from these transmission spectra is that the presence of the QDs layer does not significantly affect the self-assembly process and the optical performance of the metasurface; the increased intensity of the localized plasmonic response around 400 nm, and some spectral difference between the samples with (black curve) and without (blue curve) QDs layer, can be attributed to spot-to-spot variation within the same sample. One of the fundamental properties of the described SLRs is their diffractive nature, which can still be described with simple Bragg conditions using an effective refractive index taking into account the presence of both glass and SU8. Moreover, in order to correctly identify the

R-W's anomalies and experimentally verify their angular dispersion, it is fundamental to account for the dependency on the azimuthal angle (i.e., the orientation of the sample in the plane perpendicular to the normal direction of the illumination), an effect introduced by the two-dimensionality of the fabricated metasurfaces. For a square lattice array^[30]

$$1 - \left[\left(\frac{m\lambda_{RW}}{n_{eff}\Lambda} - \frac{n_{air}}{n_{eff}} \sin \theta_{inc} \cos \varphi_{inc} \right)^2 + \left(\frac{l\lambda_{RW}}{n_{eff}\Lambda} - \frac{n_{air}}{n_{eff}} \sin \theta_{inc} \sin \varphi_{inc} \right)^2 \right] = 0 \quad (1)$$

where Λ is the lattice period, n_{eff} is the effective refractive index, θ is the illumination angle with respect to the normal direction, φ is the azimuthal angle, and m and l are integers indicating the diffraction orders. The transmitted output light by the designed photonic grating is subjected to Snell's law. In order to experimentally verify this behavior, a customized optical setup was designed to explore the angular dispersion of the transmission spectra of the fabricated metasurfaces.^[30] The resulting contour plots in Figure 2b,c report a clear dependence on the illumination angle, confirming the diffractive nature of the main optical feature centered at 611 nm, showing an excellent agreement with the behavior predicted by Equation (1). This highly sensitive excitation angle-dependent SLR enables a finer and continuous tuning of the resonance wavelength, something that cannot be achieved by simply altering the nanoparticle composition, enabling the detailed investigation of the effect of the SLR when interacting with a quantum emitter.

Once the optical response of the fabricated substrates was characterized, confirming the spectral match between the SLR and the QDs emission, we moved forward, focusing our attention on the emission properties of our architecture. Similarly to the case of the transmission measurements, the effect of SLRs on the QDs emission behavior can be better examined by measuring the emission peak as a function of the excitation angles. First, we evaluated the photoluminescence of QD thin films without any plasmonic cavity by irradiating the samples with 515 and 1030 nm pulsed lasers to perform the 1P and 2P excitation experiments, respectively. This provided us with a

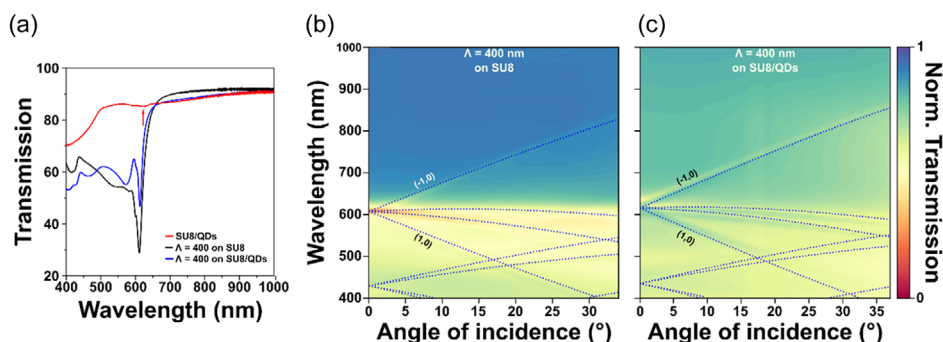


Figure 2. a) Transmission spectra of a layer of SU8/QDs in the absence (red) and presence (blue and black) of AgNPs array. The red arrow indicates the characteristic absorption peak of the QDs that can be observed in Figure S1b, Supporting Information. b,c) Contour plots showing the angular dependence of the diffractive grating assembled on top of (b) SU8 alone and (c) SU8/QDs layer from the illumination angle. The black dashed lines represent the fitted diffraction orders using Equation (1).

reference point when trying to elucidate the role of the plasmonic array (a description of the optical setup can be found in the **Experimental Section**). The 2P excitation at 1030 nm corresponds to the fundamental output wavelength of the laser, guaranteeing maximum excitation power. As expected, the QDs alone do not show any angular dependence in their fluorescence for both excitation wavelengths, with an invariant emission band centered at 618 nm (Figure S6 and S7, Supporting Information). In the presence of the plasmonic array, under 1P pumping (515 nm), a strong modulation of the emission is observed (Figure 3a). At normal incidence, the emission appears broader (Gaussian full width at half maximum (FWHM) of 29.8 ± 0.2 nm) and blue-shifts, with a maximum peak at 609 nm (black solid line in Figure 3b). This blueshift has been interpreted as a consequence of the interaction of the excitonic states with the plasmonic structure.^[47,52–54] When the plasmon resonance is close to the gain media emission wavelength, an enhanced emission efficiency is expected, due to the increased local density of photonic states around metal nanocrystals, although the effects of excitation and emission enhancement are difficult to separate.^[55,56] In the fabricated metasurfaces, both LSPRs and SLRs overlap with the absorption and emission wavelengths of the QDs. This dual-resonance condition, accompanied by the correlated EM field enhancement, leads to a complex modification of the excitonic coupling and alteration of the emission dynamics of the QDs. The plasmonic environment modifies the energy distribution of the QDs emission, influencing which excitonic states favorably emit, while the local EM field enhancement by plasmonic NP can increase QDs excitation efficiency.^[54] Taking these considerations into account, the observed blueshift can be attributed to the plasmon-induced modification of excitonic states, where the increased local density of states and altered radiative recombination pathways could favor emission from higher-energy excitonic states.^[57]

Moreover, the 1P emission difference spectral profile (calculated by subtracting the emission spectrum at the previous excitation angle) shows an illumination angle dependency that follows the diffraction lines of the plasmonic arrays in the angles between 0° and 8° (Figure 3c). This angle-dependent feature shows a narrower bandwidth (FWHM of 21.6 ± 0.2 nm) and a maximum emission enhancement observed around 5.0° , with an emission centered at 613 nm (Figure 3b) and an enhancement factor of ≈ 15 (more details on the enhancement factor calculations can be found in the Supporting Information). At a higher excitation angle (8.0°), the appearance of additional shoulders

was observed in the emission, which can be attributed to the interaction with other diffraction branches of the SLR arising from the metasurface (Figure 2c).

Following the characterization of the 1P emission process, the samples underwent a final set of experiments under 2P excitation (NIR, 1030 nm). At this wavelength, there is no direct 1P absorption process for the QDs and no direct near-field excitation of the AgNP arrays (Figure 2a), enabling a more straightforward exploration of the SLR-enhanced 2P excited emission process. Overall, the 2P excited emission spectra revealed a strong modulation and reshaping of the emission profile as a function of the illumination angle from 0° to 8° (Figure 4a). The first important observation is that, similarly to what we observed for the 1P excitation case, the integration of the AgNPs metasurface caused a significant blueshift of the QDs emission along the normal excitation direction compared to the QDs thin film alone, going from 618 nm (Figure S7, Supporting Information) to 612 nm (black solid line in Figure 4b).

The emission intensity map at different excitation powers shows a good correlation (Figure S8, Supporting Information), indicating no observable excitation power dependence on the emission energy. Nonetheless, a significantly higher signal-to-noise ratio is reached at lower excitation power in the presence of the plasmonic array compared to the cases without the plasmonic structure, and the emission peak wavelength traces the modulation of the first-order SLR with the illumination angle. Moreover, the difference spectral map (Figure 4c) shows a less pronounced SLR crossing the QDs emission peak compared to the 1P case since the SLR was not directly excited by the NIR excitation line. These observations combined suggest an emission enhancement due to the plasmon–exciton coupling. At a higher excitation power, the decrease in overall emission intensity and the absence of any emission spectral shifts at the maximum overlapping angle (2° – 5°) indicate that the sample degraded under intense EM fields (Figure S9, Supporting Information).

The 2P emission enhancement factor is estimated to be $\approx 50\times$, although we expect this value to be significantly underestimated since the actual spatial distribution of the delocalized SLR at a given time can only be approximated (see Supporting Information for more details on the enhancement factor calculations). Although the spatial distribution of the electric field could potentially be measured experimentally, this would require highly specialized equipment, and it has not been demonstrated on a delocalized SLR system.^[58] In order to circumvent these

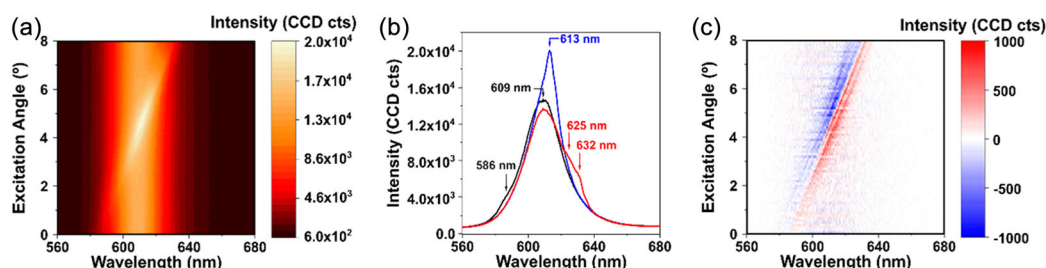


Figure 3. a) Intensity map displaying the 1P emission angular characterization of a sample made with a plasmonic array (lattice pitch of 400 nm) assembled on top of a layer of QDs (excitation wavelength 515 nm, excitation power 1.0 mW, acquisition time 5 ms). b) 1P Emission spectra at 0.0° (black), 5.0° (blue), and 8.0° (red). c) 1P emission difference spectral profile highlighting the diffraction mode of the plasmonic metasurface.

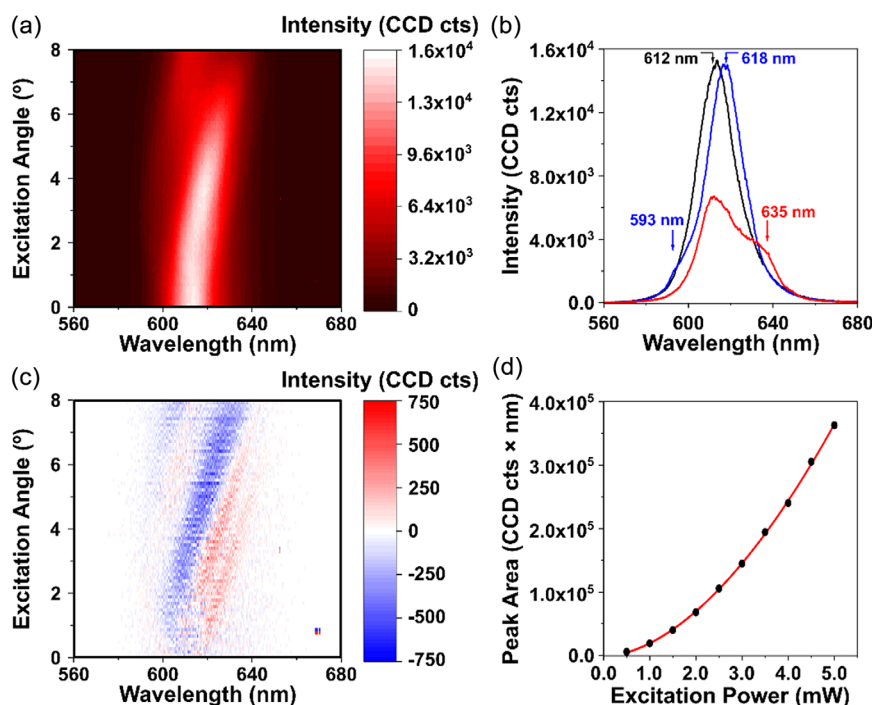


Figure 4. a) Intensity map displaying the 2P emission angular characterization of a sample made with a plasmonic array (lattice pitch of 400 nm) assembled on top of a layer of QDs (excitation wavelength 1030 nm, excitation power 5.0 mW, acquisition time 10 s). b) 2P Emission spectra at 0.0° (black), 3.2° (blue), and 8.0° (red). c) 2P emission difference spectral profile highlighting the diffraction mode of the plasmonic metasurface. d) 2P emission power dependence evaluated at 3.2° plotted with a power-law fitted line (red).

limitations, we evaluated the power exponent for the QDs in solution, the QDs thin film on SU8, and the completed architecture (Figure 4d and S11, Supporting Information and Table 1) by fitting the collected data with a power law function ($y = a + bx^c$). Our results show a clear superlinear dependency at a selected excitation angle of 3.2° (whereas 0° implies normal incidence).

Due to the quadratic dependence of the 2P absorption process ($I_{\text{ems}} \propto I_{\text{inc}}^2$), 2P excited emission is expected to show a quadratic behavior as a function of the excitation power.^[31,32,34,35,59–61] As shown in Table 1, the presence of the plasmonic metasurface induces a significant increment in the exponent value of the power dependence of the QDs 2P emission, an expected consequence of the plasmon–exciton interaction. However, although the presence of the AgNP metasurface yields a significant enhancement, the system did not reach the expected quadratic trend. This discrepancy can be partly attributed to the energy absorbed and lost in nonradiative decays of the photoexcited carriers, such as Auger recombination and exciton reabsorption.^[62,63] This hypothesis is supported by the low exponent associated with the 2P emission of the QDs in solution and inside the thin film. These nonradiative decay pathways could

be minimized utilizing type II core–shell QDs, for which the spatial separation of the carriers reduces these interactions, improving the radiative efficiency of the upconversion process.^[64–68]

3. Discussion

In order to gain a fundamental understanding of the plasmonic-enhanced 2P-excited emission, it is important to break down the 2P absorption and emission as separate processes. The presence of the plasmon cavity can increase the efficiency of either process if the cavity mode overlaps one of the two bands (absorption or emission). In the first case, the localized electric field enhancement resulting from plasmon resonances can increase the effective cross section for 2P absorption, facilitating the promotion of electrons to a higher energy level. In the case of emission, the enhanced local EM field can influence the radiative decay rate of the excited state, potentially leading to enhanced emission rates or altered emission spectra compared to conventional fluorescence processes. Since no plasmonic activity matches the excitation used for the 2P process, we assume the fabricated metasurfaces do not provide any 2P absorption enhancement.

Comparing the profile at 0° of the 1P and 2P case (black curves in Figure 3b and Figure 4b), in the first case, the spectral profile appears broadened, probably due to the excitement of localized resonances of the clusters, which in the 2P case is not directly excited. Another possible explanation would imply that a second harmonic generation signal of the plasmonic array or the QDs affects the radiative decay pathway of the QDs. This possibility

Table 1. Exponent values for the 2P excited samples.

AgNPs array on QDs/SU8	1.79 ± 0.03	$R^2 = 1.00$
QDs/SU8	1.45 ± 0.01	$R^2 = 0.99$
QDs in solution	1.38 ± 0.01	$R^2 = 0.99$

was ruled out since there is no observed second harmonic generation from the metasurface (Figure S10, Supporting Information).

Moving our attention to the observed spectral reshaping as a function of the illumination angle, it can be the result of competing and coexisting phenomena, including the modulation of far-field spontaneous emission through Purcell enhancement via coupling with a plasmonic structure^[69,70] and/or waveguiding effects that led to the suppression of scattering optical losses at certain angles. Additionally, the QD deposition onto the SU8 film can produce heterogeneity that influences the 2P photoluminescence process, while the thickness of the QDs film can change the effective refractive index of the waveguide mode, and this allows the wavelengths of the SLR modes to be controlled.^[9] As a result, the specific order of the lattice mode near the Γ point (0°) in the lattice structure shifts their wavelengths to match the emission range of the QDs. For this reason, inhomogeneity on the film thickness may cause the variation of the SLR cavity modes selection for promoting the emission, which can lead to the enhancement of certain wavelengths or directions of the emitted light. The shape of the emission reflects the interaction among plasmonic NP within the lattice. For certain angles and directions (high-symmetry points), the emission can display different branches due to the coupling with specific diffraction modes of the SLR modes, such as $(\pm 1, 0)$ and $(0, \pm 1)$.^[30,71] Finally, the potential of plasmonic enhancements using ultrafast laser pulses could be inherently affected by thermal reshaping effects on the structures and the transient broadening of plasmon resonances.^[72] Despite the significant advantages offered by plasmonic enhancement, these conditions impose some constraints on the usable power of the laser pulses, which can potentially hinder practical applications.

In summary, we report a study on the nonlinear optical effects displayed by a colloidal-based plasmonic metasurface architecture in contact with a highly photostable gain media. The results highlight an enhancement of emission processes of a QDs thin film, due to the EM enhancement associated with the local and delocalized plasmonic resonances arising from our plasmonic open optical cavity. Moreover, the diffractive nature of the SLRs enables an angular dependence of the emission properties, by showing significant alteration in the profile and a spectral modulation of ≈ 30 nm. A small deviation from the typical quadratic dependence of emission intensity on excitation was observed. This discrepancy can be attributed to the fact that the optimal quantum yield for the 2P-excited emission process of the QDs is not reached, and part of the absorbed energy is dissipated through nonradiative pathways and plasmon–phonon coupling. Fundamentally, these nonradiative pathways and plasmonic couplings (Schematic 1) are important to identify in order to gain a full mechanistic understanding of the delocalized SLR contributions. Experimentally, one can envision ultrafast pump–probe spectroscopies with the emerging attosecond radiation that could be applied to fill the current knowledge gap once these techniques are mature. Additionally, new self-consistent numerical simulations, such as recently developed semiclassical methods,^[73] could potentially unravel the underlying energy couplings and quantitatively predict the power law exponents. However, this method still needs further development to account for the nonradiative processes. Nonetheless, we observe a favorable

signal-to-noise ratio at lower excitation powers in the presence of the array, due to the plasmon–exciton coupling, which is translated into a strong enhancement of 2P-excited emission. Our results highlight the potential of colloidal-based plasmonic metasurfaces as excellent, versatile, and easily manipulable photonic crystals for exploring applications based on 2P-excited fluorescence.

In the future, to achieve more efficient coupling of the QDs excitons to this kind of plasmonic nanostructure, one could envision a fabrication scheme that assembles well-mixed QDs and plasmonic NP, optimizing the colloidal ratio, distribution, and separation to minimize optical losses. However, this represents a significant challenge, due to the different solvents and stabilizing agents that are typically involved in the preparation of plasmonic NPs and QDs. Additionally, using type-II core–shell QDs in place of the type-I ones could reduce the Auger recombination and promote the realization of a more efficient nonlinear nanophotonic device.^[64,74] Another interesting approach to improve the quality factor of the plasmonic modes could be the introduction of a tunable superstrate that can provide additional control over the effective refractive index of the cavity, thereby influencing both the resonance condition and enhancing the field confinement. These achievements marked a significant stepping stone for the use of colloidal systems in complex plasmonics and photonics materials, opening new and exciting possibilities for dynamically tuning the optical properties of the metasurfaces through their interaction with the chemical environment.

4. Experimental Section

Materials: For the synthesis of silver NP, the used reagents were silver nitrate (AgNO_3 ; 99%, CAS: 7761-88-8); sodium citrate tribasic dihydrated (SC, $\geq 99.0\%$, CAS: 6132-04-3); tannic acid (TA, CAS: 1401-55-4); poly(ethylene glycol) methyl ether thiol (PEG, MW 2000 g mol⁻¹); hexadecyltrimethylammonium chloride (CTAC, 25 wt% in water, CAS: 112-02-7) and were purchased from Sigma Aldrich. For the silanization of the masters, trichloro (1H,1H,2H,2H-perfluorooctyl) silane (97%, CAS: 78 560-45-9) was purchased from Sigma Aldrich. Polydimethylsiloxane (PDMS, Sylgard 184) used for the stamps was purchased from Dow Corning (Michigan, USA). The hard PDMS (h-PDMS) was prepared by components purchased from Gelest (CymitQuimica, Spain): (7.0–8.0% vinylmethylsiloxane)-dimethylsiloxane copolymer, trimethylsiloxy terminated (CAS: 67 762-94-1), 1,3,5,7-tetramethylcyclotetrasiloxane (95%, CAS: 2370-88-9), platinum-divinyltetramethyldisiloxane complex (2% in Pt in xylene, CAS: 68 478-92-2), and (25–35% methylhydrosiloxane)-dimethylsiloxane copolymer, trimethylsiloxane terminated (CAS: 68 037-59-2). The glass substrates were cleaned using a piranha solution prepared by mixing 95–98% w w⁻¹ sulfuric acid (CAS: 7664-93-9) and 30% w w⁻¹ hydrogen peroxide (CAS: 7722-84-1) in a 3:1 ratio, purchased from Fisher Scientific. CdSe/ZnS core–shell QDs (quantum yield: $\approx 50\%$),^[75] with an emission band centered at 620 nm and stabilized with octadecylamine ligands, were purchased from Sigma Aldrich (QDs). The negative photoresist SU8-2000.5 (14% in thinner) was purchased from KAYAKU Advanced Materials (Westborough, MA 01581). The SU8-2000 thinner was purchased from MicroCHEM (Westborough, MA 01581).

Synthesis of Silver Nanoparticles: Silver NP (AgNPs) with an average diameter of 18 ± 2 nm were synthesized combining citrate and tannic acid, adapting a protocol originally reported by Bastus et al.^[76] Three stock solutions were prepared: silver nitrate (AgNO_3) 25, sodium citrate (SC) 25, and tannic acid (TA) 2.5 mM. Next, 200 mL of an aqueous solution of SC (5 mM) and 2 mL of the TA stock solution were mixed under vigorous stirring in a three-neck round-bottomed flask. The synthetic setup was completed by a silicon oil bath, a condenser, and a digital thermometer placed

in the oil bath. The temperature was raised until vigorous boiling, after which 2 mL of the AgNO_3 stock solution was added into the solution, causing the color of the growth mixture to change from a faint yellow (caused by the presence of TA) to the characteristic dark brown coloration associated with the formation of Ag seeds of ≈ 10 nm in diameter (LSPR maximum at 402 nm). After 5 min, the formation of the Ag seeds was completed, and the temperature was lowered to 90 °C, after which 200 μL of SC, 500 μL of TA, and 500 μL of AgNO_3 stocks were sequentially injected into the growth mixture, with a delay time of ≈ 1 min after each injection. After 15 min of growth, an aliquot of 1 mL was extracted for UV vis characterizations to estimate the average AgNP diameter. Typically, the particle would reach the desired dimension in a single growth step.

N.B.: If this was not the case, more growth steps can be repeated until the desired dimension was achieved, adding 200 μL of SC, 500 μL of TA, and 500 μL of AgNO_3 stocks.

The obtained AgNPs were centrifuged three times (15 000 rpm for 30 min) to eliminate all unreacted species and to concentrate the colloid from 200 mL to less than 2 mL. At this point, AgNPs were coated with poly(ethylene glycol)methylether thiol (PEG) with an average molecular weight of 2000 g mol^{-1} . The PEG was weighed to reach a final concentration of 1 mg mL^{-1} in the colloidal suspension, dissolved in 150 μL of Milli-Q water, and finally added dropwise under vigorous stirring to the concentrated AgNPs colloid. The mixture was stirred overnight at room temperature to ensure complete functionalization of the surface by the thiolated polymer. In the final step, the excess of PEG was removed with two centrifugation steps (15 000 rpm for 30 min), and the colloid was resuspended in a 6:4 mixture of Milli-Q water and ethanol containing 50 μM of hexadecyltrimethylammonium chloride (CTAC).^[50]

Replica of the Original Master by Ormstamp: Ormstamp replicas were obtained directly from the original silicon masters (ThunderNIL, Italy), presenting arrays of cylindrical holes with a diameter of 200 nm, a depth of 300 nm, and a lattice period of 400 nm over a 49 mm^2 area. The silicon masters were coated with perfluorooctyl-trichlorosilane via chemical vapor deposition under vacuum for 30 min. This treatment prevented the Ormstamp to stick permanently to the silicon surface of the masters. After a cleaning step (immersion in an acetone bath at 120 °C for 20 min), a drop of Ormstamp photoresist was placed directly on top of the silanized silicon master, and a cleaned glass slide, pretreated with UV O_3 for 20 min, was gently superimposed on the master, making sure that no bubbles remain trapped between the glass and the Ormstamp layer. The photoresist was then activated by exposure to direct UV light for 10 min. Finally, crosslinking was favored by placing the substrates on a hot plate and applying a temperature ramp from 60 to 150 °C (6 °C min^{-1}) for 30 min. The process yielded Ormstamp pillar arrays (negative master) that can be used to prepare replicas of the original masters by nanoimprint lithography (operative masters, see below).

Preparation of s-PDMS Molds: s-PDMS molds were used for all AgNP self-assembly, exploiting their flexibility and durability to perform between 10 and 15 preparations using the same mold. The s-PDMS mixture was prepared by combining a 10:1 ratio of monomer and curing agent in a 50 mL falcon tube, vigorously mixing the two components for several minutes to ensure their homogeneous distribution. To remove the air bubbles, the tubes were centrifuged at 3200 rpm for 5 min. Finally, the mixture was poured onto the silanized operative masters (pillars). After curing overnight in the oven at 60 °C it was possible to proceed with demolding, resulting in a cylindrical nanoholes patterned s-PDMS.

Preparation of the SU8 and SU8/QDs Substrates: The SU8 thin layer was applied on a 20 $\times$ 200 mm cleaned glass coverslip by spin coating (two steps receipt: 500 rpm, 100 rpm s^{-1} , 10 s followed by 2000 rpm, 300 rpm s^{-1} , 30 s). Samples were first prebaked at 60 °C for 2 min and 95 °C for 5 min to eliminate any solvent residues, then exposed for 10 min to UV light, and finally hard baked at 120 °C for 20 min. The final thickness of the SU8 layer was measured to be 250 nm, confirmed by ellipsometry.

For the SU8/QDs substrates, the formation of the SU8 layer was followed by the QDs deposition. First, a QDs 0.1M toluene stock solution was prepared. This concentration was optimized to maximize the concentration of the nanocrystals and ensure, at the same time, the coverage of the entire glass coverslip. The stock solutions were always prepared fresh

for the preparation of the samples, preventing any change in concentration induced by solvent evaporation. Then, the QDs were spin coated onto the cured SU8 layer (1000 rpm, 100 rpm s^{-1} , 30 s), reaching a thickness of about 10 nm, as estimated by AFM (Figure S5, Supporting Information).

Template-Assisted Self-Assembly of AgNPs: A 1 μL drop of AgNPs@PEG was placed in the center of the substrate, followed immediately by the patterned s-PDMS mold. The self-assembly of a plasmonic nanocluster array was simply promoted by solvent evaporation. The close packing of the particles within each cluster was obtained by the optimization of several parameters, including shape and size of the cylindrical holes, nanoparticle concentration, solvent composition, and the CTAC concentration. After drying overnight, the s-PDMS mold was ready to be carefully removed.

Angular Displacement with Optical Measurements: An automated custom optical setup was designed to vary the light incident angle. A broadband laser-driven light source (EQ99X by Energetiq) was used to illuminate the sample: the light was initially collimated using an off-axis parabolic mirror ($f = 50.8$ mm, Thorlabs, MPD229-P01) and then focused on the sample using a $f = 100.00$ mm achromatic doublet lens. After interacting with the sample, the light was collected with a $f = 50.00$ mm achromatic doublet lens and refocused into a fiber-coupled spectrophotometer (Ocean Optics, Maya 2000Pro) with a range of detection 380–1200 nm. The sample holder was anchored on a rotational stage with resonant piezoelectric motors (Thorlabs, ELL18/M). This allowed the automatic rotation of the illumination angle θ (from 0° to 35°) of the sample, with a resolution of 0.1°. The sample was mounted vertically and oriented along the high-symmetry direction of the array.

Emission Characterization: The 1P and 2P photoluminescence experiments were performed using a spectroscopic system equipped with a 250 fs, 20 W, 1030 nm NIR pulsed laser (Light Conversion, Pharos). The signals were acquired in transmission geometry with a 30 cm spectrograph (Princeton Instruments, SP300i) coupled with a thermoelectric cooled charge-coupled device (CCD, Princeton Instruments, PIXIS-BR). In the case of the 2P excitation, the laser ran with a 100 kHz repetition rate, and two short pass filters at 1 μm (>OD5, Thorlabs, FESH1000) and 720 nm (OD7, Semrock, FF01-720) were used to cut the excitation beam from detection. For the 1P excitation case, the 1030 nm laser was frequency-doubled to 515 nm with a BaB_2O_4 crystal to achieve enough power in the output. Additionally, a tunable long pass filter (OD7, Semrock, Versachrome TBP01-628) was used to cut off the excitation from detection. A 90° off-axis parabolic mirror (focal length 3 inches) focused the excitation beams onto the sample. The emission signals were collected through a 1-inch aspherical lens with a focal length of 2 inches. The samples were mounted on a resonant piezoelectric rotational stage (Thorlabs, ELL18/M). The laser spot sizes were characterized using a CCD camera beam profiler (Thorlabs, BC106-VIS), measuring 215 $\times$ 215 and 231 $\times$ 231 μm for 515 and 1030 nm excitation lines, respectively. All laser power reported was measured by a thermopile power sensor (Ophir-Spiricon, 30 A-BB-18).

Electron Microscopy Characterization: SEM images were collected with a Phenom Pharos G2 Desktop FEG-SEM (ThermoScientific), operating between 5 and 15 kV in low vacuum (60 Pa). Transmission electron microscopy images were collected on carbon-coated 400 square mesh copper grids using the same instrument utilizing the dedicated scanning transmission electron microscopy detector and sample holder and operating the instrument at 20 kV.

Atomic Force Microscopy Characterization: AFM was collected with a Bruker Multimode AFM controlled with a Nanoscope V controller (Bruker), operating under Peakforce tapping mode.

Ellipsometry Characterization: Ellipsometry was used to characterize the spin-coated SU8 waveguide layer, measured with a Stokes laser ellipsometer (Gaertner Scientific Co., LSE) with a 632.8 nm laser at 70° from the sample normal.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Ylli Conti: conceptualization (equal); data curation (lead); investigation (lead); methodology (equal); writing—original draft (equal); writing—review editing (equal). **Xing He:** data curation (supporting); investigation (supporting); methodology (equal); writing—review editing (supporting). **Yen-Chen Chen:** data curation (supporting); investigation (supporting); writing—review editing (supporting). **Naihao Chiang:** conceptualization (equal); data curation (equal); formal analysis (lead); investigation (equal); writing—original draft (supporting); writing—review editing (equal). **Leonardo Scarabelli:** conceptualization (equal); data curation (supporting); methodology (equal); supervision (equal); validation (equal); writing—original draft (lead); writing—review editing (lead).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

colloidal plasmonic metasurfaces, lattice plasmon resonances, silver nanoparticles, template-assisted self-assemblies, two-photon absorption

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